

Calculus II

Simplify $\arccos(\cos(x))$

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Example

Find $\arccos(\cos 4)$.

- 4 is not between 0 and π .
- We need the angle a between 0 and π for which $\cos 4 = \cos a$.

$$\begin{aligned} a &= \pi - (4 - \pi) = 2\pi - 4 \\ \text{Therefore } \arccos(\cos 4) &= \arccos(\cos a) \\ &= a = 2\pi - 4. \end{aligned}$$